

Topology PhD Exam, May 2006

Work the following problems and show all work. Support all statements to the best of your ability.

1. Can a compact space contain a discrete infinite subspace?
2. Prove that the product of two connected spaces is connected.
3. Give an example of a space which is path connected but not locally connected.
4. Show that the Klein bottle with a deleted closed disk $K \setminus D$ is not a retract of K .
5. Prove that the group $SO(n)$ is connected in the matrix topology.

Answer the following with complete definitions or statements or short proofs.

6. State the Lefschetz fixed point theorem.
7. Construct a map $f : S^3 \rightarrow S^3$ of degree 3.
8. State the Simplicial Approximation theorem.
9. Prove that for a path connected topological group G the Euler characteristic $\chi(G)$ is zero.
10. Does every function $f : \mathbb{N} \rightarrow \mathbb{R}$ admit a continuous extension $\bar{f} : \beta\mathbb{N} \rightarrow \mathbb{R}$ to the Stone–Čech compactification?
11. What can you say about the k -homology groups of a closed orientable n -manifold for
 - (a) $k = n$?
 - (b) $k = n - 1$?
12. Give a definition of the compact-open topology for the space of functions.
13. Let $j : S^1 \subset S^4$ be an embedding. Can $\pi_1(S^4 \setminus j(S^1))$ be trivial? If it is not trivial, can the fundamental group $\pi_1(S^4 \setminus j(S^1))$ be abelian?
14. Give a definition of the wedge of a family of topological spaces. Prove that an infinite wedge of circles is not metrizable.
15. State the Contraction Mapping Theorem.